Diseño de Algoritmos – Práctica 0

Exercises about GNU gprof

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Exercises

Exercise 1

In the program below, the first terms of the *Fibonacci sequence* are computed by two different functions defined in separate modules:

```c
#include <stdio.h>
#include <stdlib.h>

#include "fibonacci.h"

int main(int argc, char *argv[])
{
  if (argc != 2) {
    fprintf(stderr, "Usage: %s n\n", argv[0]);
    return 1;
  }
  natural n = atol(argv[1]);
  printf("Fibonacci numbers (n = %lu)\n", n);
  puts("\nIterative algorithm:\n");
  for (natural k = 0; k <= n; ++k)
    printf("F1(%lu) = %lu\n", k, f1(k));
  puts("\nRecursive algorithm:\n");
  for (natural k = 0; k <= n; ++k)
    printf("F2(%lu) = %lu\n", k, f2(k));
  return 0;
}
```

File 1: `main.c`

```c
typedef unsigned long int natural;

natural f1(natural n);
natural f2(natural n);
```

File 2: `fibonacci.h`

```c
#include "fibonacci.h"

natural f1(natural n)
{
  natural a = 0, b = 1;
  while (n > 1) {
    const natural c = a + b;
    a = b;
    b = c;
    --n;
  }
  return n ? b : a;
}
```

File 3: `fibonacci-iterative.c`

```c
#include "fibonacci.h"

natural f2(natural n)
{
  if (n < 2)
    return n;
  else
    return f2(n - 1) + f2(n - 2);
}
```

File 4: `fibonacci-recursive.c`

Compile separately and link the modules into a program for profiling. Run the program twice, with 25 and 45 as command-line parameters, respectively.[1] For each run, answer the following questions, remarking any significant difference that you may find between both runs:

1. Generate the flat profile and the call graph in separate files.
2. How long does it take to execute the whole program?
3. How long does it take to execute each function?
4. How many times is each function invoked?
5. How many recursive calls are made? What functions are involved?
6. List the parents and children for each function and draw the call graph *by hand*.
7. Generate and export to PDF the *full* visual call graph with the appropriate tools.

[1] This could take up to a few minutes.